

CBSE Class 10 Mathematics — Statistics — Solutions

1. 1. The mean of the first 5 natural numbers is:

Answer: (B)

Solution:

1. Identify the first 5 natural numbers: 1, 2, 3, 4, 5.
2. Sum the numbers: $1 + 2 + 3 + 4 + 5 = 15$.
3. Divide the sum by the count of numbers: $15 / 5 = 3$.

Derivation:

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

Statistics — Mean

2. 2. Find the mode of the following data: 2, 3, 4, 3, 5, 3, 6, 3.

Answer: (C)

Solution:

1. Observe the frequency of each number.
2. The number 3 appears four times, which is more frequent than any other number.
3. Therefore, the mode is 3.

Derivation:

Mode is the observation with the highest frequency. Since 3 occurs 4 times, Mode = 3.

Statistics — Mode

3. 3. If the mean of $x, x + 2, x + 4, x + 6, x + 8$ is 24, find the value of x .

Answer: (A)

Solution:

1. Sum the observations: $x + (x+2) + (x+4) + (x+6) + (x+8) = 5x + 20$.
2. Divide by number of observations (5) and equate to mean (24).
3. Solve for x : $(5x + 20) / 5 = 24 \Rightarrow x + 4 = 24 \Rightarrow x = 20$.

Derivation:

$$\begin{aligned}\frac{5x + 20}{5} &= 24 \\ \Rightarrow x + 4 &= 24 \\ \Rightarrow x &= 20\end{aligned}$$

Statistics — Mean

4. **4.** The median of first 7 prime numbers is:

Answer: (B)

Solution:

1. List the first 7 prime numbers: 2, 3, 5, 7, 11, 13, 17.
2. Since $n=7$ (odd), the median is the $((n+1)/2)$ th observation.
3. The 4th observation is 7.

Derivation:

$$\text{Median} = \left(\frac{7+1}{2} \right) \text{th term} = 4\text{th term} = 7$$

Statistics — Median

5. **5.** For a given distribution, if Mean = 20 and Mode = 20, find the Median using the empirical relationship.

Answer: (C)

Solution:

1. Use the empirical formula: $\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$.
2. Substitute the given values: $20 = 3(\text{Median}) - 2(20)$.
3. $20 = 3(\text{Median}) - 40 \Rightarrow 60 = 3(\text{Median}) \Rightarrow \text{Median} = 20$.

Derivation:

$$\begin{aligned} 20 &= 3(\text{Median}) - 2(20) \\ \Rightarrow 20 + 40 &= 3(\text{Median}) \\ \Rightarrow \text{Median} &= 60/3 = 20 \end{aligned}$$

Statistics — Empirical Relationship

6. **6.** Assertion (A): The mean of the first five natural numbers is 3. Reason (R): The mean of a set of observations is the sum of observations divided by the total number of observations.

Answer: (A)

Solution:

1. $\text{Sum of first 5 natural numbers} = 1 + 2 + 3 + 4 + 5 = 15$
2. $\text{Mean} = \text{Sum} / \text{Number of observations} = 15/5 = 3$
3. Assertion is true and Reason is the correct definition/formula.

Derivation:

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

Statistics — Mean of ungrouped data

7. **7.** Assertion (A): The empirical relationship between mean, median, and mode is $3\text{Median} = \text{Mode} + 2\text{Mean}$. Reason (R): For a symmetric distribution, mean, median, and mode are equal.

Answer: (A)

Solution:

1. The empirical formula is indeed $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$.
2. In a perfectly symmetric distribution, $\text{mean} = \text{median} = \text{mode}$, which satisfies the formula.
3. Both statements are true and related to the central tendency properties.

Derivation:

$$3\text{Median} = \text{Mode} + 2\text{Mean}$$

Statistics — Relationship between mean, median and mode

8. **8.** Assertion (A): The mode of the data 2, 3, 4, 3, 2, 3 is 3. Reason (R): Mode is the observation that occurs most frequently.

Answer: (A)

Solution:

1. Count frequencies: 2 appears twice, 3 appears three times, 4 appears once.
2. The most frequent value is 3.
3. Assertion is true and Reason is the correct definition of mode.

Derivation:

$$\text{Frequency of 2} = 2, \text{Frequency of 3} = 3, \text{Frequency of 4} = 1. \therefore \text{Mode} = 3$$

Statistics — Mode of ungrouped data

9. **9.** Assertion(A) : If the mean of a distribution is 20 and the sum of observations is 100, then the number of observations is 5. Reason(R) : The sum of observations = Mean $\times$ Number of observations.

Answer: (A)

Solution:

1. Given Mean = 20, Sum = 100.
2. Using formula: $100 = 20 \times n \Rightarrow n = 100 / 20 = 5$.
3. Assertion is true and Reason provides the correct formula used for calculation.

Derivation:

$$n = \frac{\sum x_i}{\bar{x}} = \frac{100}{20} = 5$$

Statistics — Mean of ungrouped data

10. **10.** Assertion (A): The median of the data 5, 8, 12, 16, 20 is 12. Reason (R): The median is always the middle value when the data is arranged in ascending order.

Answer: (A)

Solution:

1. Data: 5, 8, 12, 16, 20 (already in ascending order).
2. There are 5 observations (odd). Median is the $(n+1)/2$ th term.
3. 3rd term is 12. Assertion is true and Reason is the correct definition.

Derivation:

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term} = \left(\frac{5+1}{2} \right)^{\text{th}} = 3^{\text{rd}} \text{ term} = 12$$

Statistics — Median of ungrouped data

- 11. 11.** The mean of 5 observations is 15. If each observation is increased by 3, find the new mean.

Answer:

18

Solution:

1. Use the property that if each observation is increased by a constant k, the mean also increases by k.
2. Add the constant to the original mean.

Derivation:

$$\text{Original Mean } (\bar{x}) = 15. \text{ New Mean } = \bar{x} + 3 = 15 + 3 = 18.$$

Statistics — Mean of grouped/ungrouped data

- 12. 12.** Find the class mark of the class interval 150-200.

Answer:

175

Solution:

1. Identify the lower limit (L) and upper limit (U) of the interval.
2. Apply the formula: Class mark = $(L + U) / 2$.

Derivation:

$$\text{Class mark} = \frac{150 + 200}{2} = \frac{350}{2} = 175.$$

Statistics — Class intervals and class marks

- 13. 13.** If the mode of a data set is 45 and the mean is 27, find the median using the empirical relationship.

Answer:

33

Solution:

1. State the empirical formula: Mode = 3 Median - 2 Mean.
2. Substitute the given values and solve for Median.

Derivation:

$$\begin{aligned}45 &= 3(\text{Median}) - 2(27) \\ \implies 45 &= 3(\text{Median}) - 54 \\ \implies 99 &= 3(\text{Median}) \\ \implies \text{Median} &= 33.\end{aligned}$$

Statistics — Relationship between Mean, Median and Mode

- 14. 14.** The frequency of a class is 12 and the total number of observations is 40. Find the relative frequency of this class.

Answer:

$$0.3$$

Solution:

1. Use the formula for relative frequency: (Frequency of class) / (Total number of observations).
2. Calculate the division.

Derivation:

$$\text{Relative Frequency} = \frac{12}{40} = \frac{3}{10} = 0.3.$$

Statistics — Frequency distribution

- 15. 15.** Given the data: 12, 15, 18, 12, 15, 12, 18. Find the range of the data.

Answer:

$$6$$

Solution:

1. Identify the maximum and minimum values in the data set.
2. Subtract the minimum from the maximum.

Derivation:

$$\text{Max} = 18, \text{Min} = 12. \text{Range} = 18 - 12 = 6.$$

Statistics — Measures of dispersion

- 16. 16.** The mean of the following frequency distribution is 50. Find the value of p if the sum of frequencies is 100: — Class — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 —
— Frequency — 17 — p — 32 — p — 19 —

Answer:

$$p = 16$$

Solution:

1. Sum the frequencies to find the relationship between p and total frequency.
2. Calculate the class marks (x_i) for each interval.
3. Use the formula for mean $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ and solve for p .

Derivation:

$$\Rightarrow p = 16. \text{Classmar}$$

$$\Rightarrow p = 16. \text{Calculation : } 170 + 480 + 1600 + 1120 + 1710 = 5080. 5080/100 = 50.8. \text{Given Mean is}$$

Statistics — Mean of grouped data

- 17. 17.** The following distribution shows the daily pocket allowance of children of a locality. If the mean pocket allowance is Rs. 18, find the missing frequency f : —
 Daily allowance (in Rs) — 11-13 — 13-15 — 15-17 — 17-19 — 19-21 — 21-23 —
 23-25 — ————— — Number of children — 7 — 6 —
 9 — 13 — f — 5 — 4 —

Answer:

$$f = 6$$

Solution:

1. Determine the class mark (x_i) for each interval.
2. Apply the direct method formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.
3. Substitute known values and solve the resulting linear equation for f .

Derivation:

$$\text{Let } A = 18. x_i \text{ values : } 12, 14, 16, 18, 20, 22, 24. \sum f_i = 44 + f. \sum f_i x_i = 84 + 84 + 144 + 234 + 20f$$

Statistics — Mean of grouped data

- 18. 18.** Find the median of the following data: — Marks — 0-10 — 10-20 — 20-30 —
 30-40 — 40-50 — ————— — Number of students — 5 — 15 —
 20 — 7 — 3 —

Answer:

$$\text{Median} = 23.5$$

Solution:

1. Construct a cumulative frequency table.
2. Identify the median class where $N/2$ falls.
3. Use the formula $\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$.

$N = 50, N/2 = 25. Medianclass is 20-30. l = 20, cf = 20, f = 20, h = 10. Median = 20 + ((25 - 20) / 20)$

19. The following table shows the ages of 100 patients admitted in a hospital during a year. Find the mode of the ages: — Age (in years) — 5-15 — 15-25 — 25-35 — 35-45 — 45-55 — 55-65 — ————— — Number of patients — 6 — 11 — 21 — 23 — 14 — 5 —

$$Mode = 36.82 years$$

1. Identify the modal class (class with highest frequency).
2. Apply the formula $\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$.
3. Substitute values $l = 35, f_1 = 23, f_0 = 21, f_2 = 14, h = 10$.

$$\text{Mode} = 35 + \left(\frac{23 - 21}{2(23) - 21 - 14} \right) \times 10 = 35 + \left(\frac{2}{46 - 35} \right) \times 10 = 35 + (2/11) \times 10 = 35 + 1.82 = 36.82.$$

20. 20. A survey conducted on 20 households in a locality by a group of students resulted in the following frequency table for the number of family members in a household: —
Family size — 1-3 — 3-5 — 5-7 — 7-9 — 9-11 — ————— —
Frequency — 7 — 8 — 2 — 2 — 1 — Find the mode of this data.

$$Mode = 3.286$$

1. Identify the modal class as 3 – 5 since it has the highest frequency 8.
2. State the parameters: $l = 3, f_1 = 8, f_0 = 7, f_2 = 2, h = 2$.
3. Calculate using the mode formula.

$$\text{Mode} = 3 + \left(\frac{8-7}{2(8)-7-2} \right) \times 2 = 3 + \left(\frac{1}{16-9} \right) \times 2 = 3 + 2/7 = 3 + 0.286 = 3.286.$$

21. The mean of the following frequency distribution is 50. Find the values of p and q , if the sum of all frequencies is 120.

— Class Interval —	0-20	20-40	40-60	60-80	80-100	— — :	— — :	— — :
— : — — :	— — :	— — :	— — —	Frequency —	17 — p —	32 — q —	19 —	

 $p = 28, q = 24$

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1. Sum the frequencies and equate to 120 to form the first equation: $17 + p + 32 + q + 19 = 120$.
2. Simplify the equation to $p + q = 52$.
3. Calculate the mid-points (x_i) for each interval: 10, 30, 50, 70, 90.
4. Calculate $f_i x_i$: 170, $30p$, 1600, $70q$, 1710.
5. Use the mean formula $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ and substitute known values.
6. Solve the resulting linear equation $30p + 70q = 2520$ alongside $p + q = 52$ to find the values.

Derivation:

$$\begin{aligned}
 17 + p + 32 + q + 19 &= 120 \\
 \implies p + q &= 52 \quad (1) \\
 \sum f_i x_i &= 170 + 30p + 1600 + 70q + 1710 = 3480 + 30p + 70q \\
 50 &= \frac{3480 + 30p + 70q}{120} \\
 6000 &= 3480 + 30p + 70q \\
 30p + 70q &= 2520 \\
 \implies 3p + 7q &= 252 \quad (2) \\
 3(52 - q) + 7q &= 252 \\
 156 - 3q + 7q &= 252 \\
 4q &= 96 \\
 \implies q &= 24 \\
 p &= 52 - 24 = 28
 \end{aligned}$$

Statistics — Mean of Grouped Data

- 22. 22.** The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

— Class Interval —	0-100	—	100-200	—	200-300	—	300-400	—	400-500	—	500-600
—	600-700	—	700-800	—	800-900	—	900-1000	—	:	—	:
—	:	—	:	—	:	—	:	—	:	—	:
—	20	—	y	—	9	—	7	—	4	—	Frequency
									2	—	5
									x	—	12
									17		

Answer:

$$x = 9, y = 15$$

Solution:

1. Sum all frequencies to find $x + y = 24$.
2. Identify the median class using the median 525, which is 500 – 600.
3. Set up the median formula: Median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$.
4. Substitute $l = 500, n = 100, cf = 36 + x, f = 20, h = 100$.
5. Solve for x , then substitute back into the sum equation to find y .

Derivation:

$$\begin{aligned}\sum f &= 76 + x + y = 100 \\ \implies x + y &= 24 \\ 525 &= 500 + \left(\frac{50 - (36 + x)}{20} \right) \times 100 \\ 25 &= (14 - x) \times 5 \\ 5 &= 14 - x \\ \implies x &= 9 \\ y &= 24 - 9 = 15\end{aligned}$$

Statistics — Median of Grouped Data

- 23. 23.** The following distribution gives the daily income of 50 workers of a factory. Convert the distribution into a continuous form.
- | Daily Income (Rs.) | Number of Workers |
|--------------------|-------------------|
| 120-140 | 12 |
| 140-160 | 14 |
| 160-180 | 8 |
| 180-200 | 6 |
| 200-220 | 10 |

Answer:

$$\text{Median} \approx 138$$

Solution:

- Create the cumulative frequency table: less than 120 (12), 140 (26), 160 (34), 180 (40), 200 (50).
- Plot the points (120, 12), (140, 26), (160, 34), (180, 40), (200, 50) on a graph.
- Join the points with a smooth curve to get the ogive.
- Locate $N/2 = 25$ on the y-axis.
- Draw a horizontal line to intersect the ogive and drop a perpendicular to the x-axis to read the median.

Derivation:

Median is found at $n/2 = 25$ on the y-axis. The corresponding x-coordinate is approximately 138.

Statistics — Graphical Representation of Cumulative Frequency

- 24. 24.** Find the mode of the following frequency distribution. If the mean of the same data is 28.6, verify the empirical relationship between mean, median, and mode.
- | Class | Frequency |
|-------|-----------|
| 0-10 | 5 |
| 10-20 | 8 |
| 20-30 | 20 |
| 30-40 | 15 |
| 40-50 | 7 |

Answer:

$$\text{Mode} = 27.5, \text{Median} = 28.2, \text{Relationship verified as } 3 \times 28.2 - 2 \times 28.6 \approx 27.5$$

Solution:

- Identify the modal class (20 – 30) with the highest frequency (20).
- Apply the mode formula: $\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$.
- Calculate mode: $20 + \frac{20-8}{40-8-15} \times 10 = 20 + \frac{12}{17} \times 10 \approx 27.06$.

4. Calculate median using cumulative frequencies: 5, 13, 33, 48, 55.
5. Verify the formula $3 \times \text{Median} - 2 \times \text{Mean} = \text{Mode}$.

Derivation:

$$\text{Mode} = 20 + \frac{12}{17} \times 10 \approx 27.06$$

$$\text{Median} = 20 + \frac{27.5 - 13}{20} \times 10 = 20 + 7.25 = 27.25$$

$$3(27.25) - 2(28.6) = 81.75 - 57.2 = 24.55 \approx 27.06$$

Statistics — Relationship between Mean, Median, and Mode